

✓ GEOG5990M Final Assignment - RESIT

Student ID number: [202021408]

Assignment Details

This assignment should use this Jupyter Notebook to document the investigation and interpretation of the relationship between AirBNB property rentals and area characteristics (data provided) in Bristol.

A full overview of the data provided can be found below.

These data should be suitably joined and pre-processed to provide insight and two polished data visualisations produced to communicate the results of the analysis.

- The Jupyter notebook should follow the data science process outlined in the module.
- You should use markdown cells to explain and justify the steps taken in the code.
- Whilst you should use visualisations throughout for data exploration, please present two final visualisations to communicate your 'public good' insight, one spatial and one non-spatial.
- For both these visualisations indicate your intended audience (e.g. academic, the public, policymakers, business stakeholders)
- AND the subsequent data visualisation choices that have been made to ensure these outputs are suitable for this audience.

The Jupyter notebook **markdown word count** is limited to **1500 words**.

Please upload the Jupyter Notebook and all files need to run the notebook analysis as a zip file using the 'Zip file upload' option of Turnitin. Contained within the zip folder please also submit a .txt file with a link to your GitHub repository

Assessment Information

You will be assessed on the following:

- **Readability of the Jupyter notebook including markdown cells and commented code (5%)**
- **Logical Flow of the Jupyter notebook (5%)**
- **Data cleaning (30%), including:**
 - Exploratory data description (with citations where appropriate)
 - Justification of data cleaning choices (with citations where appropriate)
 - Data visualisation to aid data cleaning
- **Statistical modelling/ quantification of the association between your chosen variables (10%):**
 - Effective interpretation of the model
 - Justification of model chosen
 - Potential models include: k-means, spearman's rank, predictive model (e.g. OLS)
- **Data Visualisation of end results: (30%)**
 - This should include one spatial and one non-spatial visualisation
 - Evidence of modifying visualisations to make them:
 - Accessible (colour blind audience etc.)
 - Easy to interpret
 - Readable (e.g. change variable names from how they have been presented in original table to improve understanding)
 - Explanation of data visualisation choices (with citations where appropriate)
 - Evidence of critical selection of chosen visualisations to present (with citations where appropriate)
- **Reproducibility (20%)**
 - You should ensure all the data and steps to produce the analysis undertaken is reproducible

- You should create a **GitHub repository with suitable documentation**. This should include an introductory markdown (readme) file explaining:
 - The background/context of the project
 - What data the GitHub repository contains
 - What the code aims to do
 - Any further information someone might need to be able to run your code and reproduce your analysis
- All files needed to run the analysis
- Your .ipynb notebook

The Jupyter notebook will be marked against the SoG generic marking criteria and specifically the following additional criteria:

- Understanding of task
- Structure and flow of the report
- Breadth of content
- Reading and referencing
- Aesthetics and design
- Writing / presentation
- Data

▼ Data Overview

AirBNB data

A sample of AirBNB data for Bristol has been collected for you. AirBNB is a company operating that operates an online marketplace for short- and long-term homestays and experiences. It is often used to book holiday stays. The data provided includes...

Field	Description
id	Airbnb's unique identifier for the listing
name	
host_id	
host_name	
neighbourhood_group	The neighbourhood group as geocoded using the latitude and longitude against neighborhoods as defined by open or public digital
neighbourhood	The neighbourhood as geocoded using the latitude and longitude against neighborhoods as defined by open or public digital shape
latitude	Uses the World Geodetic System (WGS84) projection for latitude and longitude.
longitude	Uses the World Geodetic System (WGS84) projection for latitude and longitude.
price	daily price in local currency. Note, \$ sign may be used despite locale
minimum_nights	minimum number of night stay for the listing (calendar rules may be different)
number_of_reviews	The number of reviews the listing has
last_review	The date of the last/newest review
calculated_host_listings_count	The number of listings the host has in the current scrape, in the city/region geography.
availability_365	availability_x. The availability of the listing x days in the future as determined by the calendar. Note a listing may be available because
number_of_reviews_ltm	The number of reviews the listing has (in the last 12 months)
license	
property_type	Self selected property type. Hotels and Bed and Breakfasts are described as such by their hosts in this field
room_type	All homes are grouped into the following three room types: Entire place, Private room, Shared room, Hotel
accommodates	The maximum capacity of the listing
bathrooms	The number of bathrooms in the listing
bathrooms_text	The number of bathrooms in the listing. On the Airbnb web-site, the bathrooms field has evolved from a number to a textual descript
bedrooms	The number of bedrooms
beds	The number of bed(s)
amenities	

Access to Healthy Assets and Hazards (AHAH) Data

AHAH (the index of 'Access to Health Assets and Hazards') is a multi-dimensional index developed by the CDRC for Great Britain measuring how 'healthy' neighbourhoods are. The AHAH index combines indicators under four different domains of

accessibility:

- Retail environment (access to fast food outlets, pubs, tobacconists, gambling outlets),
- Health services (access to GPs, hospitals, pharmacies, dentists, leisure services),
- Physical environment (Blue Space, Green Space - Passive), and
- Air quality (NO₂, PM₁₀, SO₂).

Please look at the full AHAH metadata on the CDRC website, make sure you understand what each of the variables are showing you <https://data.geoods.ac.uk/dataset/access-to-healthy-assets-hazards-ahah>

```
# read in required packages
import pandas as pd
import geopandas as gpd
```

1.Introduce

Airbnb has become the industry leader of point-to-point accommodation platforms over the past decade and has become a model of the sharing economy [1]. For landlords, short-term tourism rentals are often more profitable than long-term residential rentals. For guests, using Airbnb allows them to experience the accommodation environment, interact with the landlord, and engage with the local culture [2]. The listings are often concentrated in regions with more abundant tourism resources [3].

When policymakers design policies, they are increasingly paying attention to the impact of geographical environment on health [4]. AHAH was created for the UK to develop a national open-source low-level geographical indicator, used to measure whether the surrounding environment of each community is beneficial to health. When AHAH was developed, 14 indicators were considered, divided into 4 categories, and the average score and comprehensive score of each neighborhood were calculated, providing reference and assistance for policymakers and researchers [5].

Bristol is one of the most popular tourist destinations in the UK, which has led to the rapid development of the sharing economy platforms in the short-term rental property market [3].

```
# read in the final project resit files

# read in AirBnB Data
airbnb =pd.read_csv('https://github.com/FrancescaPontin/GE0G5990M_final_project_resit/raw/refs/heads/main/airbnb_data_bris
# read in AHAH index data
ahah = gpd.read_file('https://github.com/FrancescaPontin/GE0G5990M_final_project_resit/raw/refs/heads/main/bristol_ahah_i
```

```
# View the first five rows of the AHAH data
ahah.head()
```

	FID	LSOA11CD	LSOA11NM	LSOA11NMW	BNG_E	BNG_N	LONG	LAT	GlobalID	lsoa11	...	ah3e_1
0	14053	E01014485	Bristol 023A	Bristol 023A	360020	174735	-2.57697	51.47015	3edee4ae- aa59-4e9e- 8066- fd81222d4d4d	E01014485	...	338
1	14054	E01014486	Bristol 023B	Bristol 023B	359924	174186	-2.57829	51.46521	756ec26e- f1c8-4911- ad30- ff9aab76e295	E01014486	...	338
2	14055	E01014487	Bristol 020A	Bristol 020A	359244	174738	-2.58814	51.47013	3b905c75- f6aa-42c6- 8347- f96f779a7b75	E01014487	...	311
3	14056	E01014488	Bristol 023C	Bristol 023C	359701	174621	-2.58155	51.46911	266e675c- 152b-4827- abb0- c9424d7b485f	E01014488	...	341
4	14057	E01014489	Bristol 023D	Bristol 023D	359346	174262	-2.58662	51.46585	b608a9d9- 8976-4047- b4c5- a5dbf541d1ac	E01014489	...	338

```
## explore the AHAH spatial data
ahah.explore()
```


	Unnamed: 0	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	reviews_per_month
0	0	70820	Townhouse in Bristol • ★4.77 • 1 bedroom • 1 bathroom • 1 bed • ...	360195	Sarah	NaN	Windmill Hill	51.43843	-2.59183	1.0
1	1	117122	Condo in Bristol • ★4.96 • 1 bedroom • 1 bed • ...	591555	Marcus	NaN	Hotwells & Harbourside	51.44976	-2.61000	1.0
2	2	176306	Townhouse in Bristol • ★4.79 • 1 bedroom • 1 bathroom • 1 bed • ...	708175	Orla	NaN	Southville	51.44174	-2.60297	1.0
3	3	307634	Townhouse in Bristol • ★4.65 • 1 bedroom • 1 bathroom • 1 bed • ...	1582810	Wendy	NaN	Easton	51.46487	-2.55690	1.0
4	4	390781	Home in Bristol • ★4.89 • 1 bedroom • 1 bed • ...	1954565	Fiona	NaN	Brislington West	51.44444	-2.55783	1.0
5 rows × 28 columns										

```
## view the columns in the Airbnb dataset
airbnb.columns

Index(['Unnamed: 0', 'id', 'name', 'host_id', 'host_name',
      'neighbourhood_group', 'neighbourhood', 'latitude', 'longitude',
      'room_type_x', 'price', 'minimum_nights', 'number_of_reviews',
      'last_review', 'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license', 'property_type',
      'room_type_y', 'accommodates', 'bathrooms', 'bathrooms_text',
      'bedrooms', 'beds', 'amenities', 'LSOAI1CD'],
      dtype='object')
```

```
# check the shape of the Airbnb dataset
airbnb.shape

(2556, 28)
```

```
# check the data types in the Airbnb dataset
airbnb.dtypes
```

0	
Unnamed: 0	int64
id	int64
name	object
host_id	int64
host_name	object
neighbourhood_group	float64
neighbourhood	object
latitude	float64
longitude	float64
room_type_x	object
price	float64
minimum_nights	int64
number_of_reviews	int64
last_review	object
reviews_per_month	float64
calculated_host_listings_count	int64
availability_365	int64
number_of_reviews_ltm	int64
license	float64
property_type	object
room_type_y	object
accommodates	int64
bathrooms	float64
bathrooms_text	object
bedrooms	float64
beds	float64
amenities	object
LSOA11CD	object

dtype: object

check the Airbnb dataset for missing data
airbnb.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2556 entries, 0 to 2555
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            2556 non-null   int64
1   id                    2556 non-null   int64
2   name                  2556 non-null   object
3   host_id               2556 non-null   int64
4   host_name             2556 non-null   object
5   neighbourhood_group    0 non-null      float64
6   neighbourhood          2556 non-null   object
7   latitude              2556 non-null   float64
8   longitude              2556 non-null   float64
9   room_type_x           2556 non-null   object
10  price                 2169 non-null   float64
11  minimum_nights         2556 non-null   int64
12  number_of_reviews      2556 non-null   int64
13  last_review            2175 non-null   object
14  reviews_per_month      2175 non-null   float64
```

```
15  calculated_host_listings_count  2556 non-null  int64
16  availability_365                2556 non-null  int64
17  number_of_reviews_ltm          2556 non-null  int64
18  license                        0 non-null     float64
19  property_type                  2556 non-null  object
20  room_type_y                    2556 non-null  object
21  accommodates                   2556 non-null  int64
22  bathrooms                      0 non-null     float64
23  bathrooms_text                 2555 non-null  object
24  bedrooms                      0 non-null     float64
25  beds                          2501 non-null  float64
26  amenities                      2556 non-null  object
27  LSOA11CD                      2555 non-null  object
```

dtypes: float64(9), int64(9), object(10)

memory usage: 559.3+ KB

```
# count missing values in the Airbnb dataset
airbnb.isnull().sum()
```

	0
Unnamed: 0	0
id	0
name	0
host_id	0
host_name	0
neighbourhood_group	2556
neighbourhood	0
latitude	0
longitude	0
room_type_x	0
price	387
minimum_nights	0
number_of_reviews	0
last_review	381
reviews_per_month	381
calculated_host_listings_count	0
availability_365	0
number_of_reviews_ltm	0
license	2556
property_type	0
room_type_y	0
accommodates	0
bathrooms	2556
bathrooms_text	1
bedrooms	2556
beds	55
amenities	0
LSOA11CD	1

dtype: int64

```
# view the Airbnb listing with missing LSOA code
airbnb.loc[airbnb['LSOA11CD'].isna()]
```


Unnamed: 0		id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude
2139	2139	930212420101313369	Home in Pill • 2 bedrooms • 3 beds • 0 baths	247416919	Sykes Holiday Cottages	NaN	Avonmouth & Lawrence Weston	51.4820
1 rows × 28 columns								

```
# view the columns in the AHAH dataset
ahah.columns

Index(['FID', 'LSOA11CD', 'LSOA11NM', 'LSOA11NMW', 'BNG_E', 'BNG_N', 'LONG',
      'LAT', 'GlobalID', 'lsoa11', 'ah3gp', 'ah3dent', 'ah3phar', 'ah3hosp',
      'ah3blue', 'ah3gpas', 'ah3ffood', 'ah3gamb', 'ah3leis', 'ah3pubs',
      'ah3tob', 'ah3no2', 'ah3so2', 'ah3pm10', 'ah3gp_rnk', 'ah3dent_rnk',
      'ah3phar_rnk', 'ah3hosp_rnk', 'ah3leis_rnk', 'ah3gpas_rnk',
      'ah3blue_rnk', 'ah3no2_rnk', 'ah3so2_rnk', 'ah3pm10_rnk', 'ah3gamb_rnk',
      'ah3pubs_rnk', 'ah3tob_rnk', 'ah3ffood_rnk', 'ah3gp_pct', 'ah3dent_pct',
      'ah3phar_pct', 'ah3hosp_pct', 'ah3leis_pct', 'ah3gpas_pct',
      'ah3blue_pct', 'ah3no2_pct', 'ah3so2_pct', 'ah3pm10_pct', 'ah3gamb_pct',
      'ah3pubs_pct', 'ah3tob_pct', 'ah3ffood_pct', 'ah3h', 'ah3g', 'ah3e',
      'ah3r', 'ah3h_rnk', 'ah3g_rnk', 'ah3e_rnk', 'ah3r_rnk', 'ah3h_pct',
      'ah3g_pct', 'ah3e_pct', 'ah3r_pct', 'ah3ahah', 'ah3ahah_rn',
      'ah3ahah_pc', 'geometry'],
      dtype='object')
```

```
# check the shape of the AHAH dataset
ahah.shape

(263, 68)
```

```
# check the data types in the AHAH dataset
ahah.dtypes
```

	0
FID	int32
LSOA11CD	object
LSOA11NM	object
LSOA11NMW	object
BNG_E	int32
...	...
ah3r_pct	int32
ah3ahah	float64
ah3ahah_rn	int32
ah3ahah_pc	int32
geometry	geometry

68 rows × 1 columns

dtype: object

```
# check the AHAH dataset for missing data
ahah.info()
```

```
<class 'geopandas.geodataframe.GeoDataFrame'>
RangeIndex: 263 entries, 0 to 262
Data columns (total 68 columns):
#   Column          Non-Null Count  Dtype
---  -
0   FID              263 non-null    int32
1   LSOA11CD         263 non-null    object
2   LSOA11NM         263 non-null    object
```

3	LSOA11NMW	263	non-null	object
4	BNG_E	263	non-null	int32
5	BNG_N	263	non-null	int32
6	LONG	263	non-null	float64
7	LAT	263	non-null	float64
8	GlobalID	263	non-null	object
9	lsoa11	263	non-null	object
10	ah3gp	263	non-null	float64
11	ah3dent	263	non-null	float64
12	ah3phar	263	non-null	float64
13	ah3hosp	263	non-null	float64
14	ah3blue	263	non-null	float64
15	ah3gpas	263	non-null	float64
16	ah3fffood	263	non-null	float64
17	ah3gamb	263	non-null	float64
18	ah3leis	263	non-null	float64
19	ah3pubs	263	non-null	float64
20	ah3tob	263	non-null	float64
21	ah3no2	263	non-null	float64
22	ah3so2	263	non-null	float64
23	ah3pm10	263	non-null	float64
24	ah3gp_rnk	263	non-null	int32
25	ah3dent_rnk	263	non-null	int32
26	ah3phar_rnk	263	non-null	int32
27	ah3hosp_rnk	263	non-null	int32
28	ah3leis_rnk	263	non-null	int32
29	ah3gpas_rnk	263	non-null	int32
30	ah3blue_rnk	263	non-null	int32
31	ah3no2_rnk	263	non-null	int32
32	ah3so2_rnk	263	non-null	int32
33	ah3pm10_rnk	263	non-null	int32
34	ah3gamb_rnk	263	non-null	int32
35	ah3pubs_rnk	263	non-null	int32
36	ah3tob_rnk	263	non-null	int32
37	ah3fffood_rnk	263	non-null	int32
38	ah3gp_pct	263	non-null	int32
39	ah3dent_pct	263	non-null	int32
40	ah3phar_pct	263	non-null	int32
41	ah3hosp_pct	263	non-null	int32
42	ah3leis_pct	263	non-null	int32
43	ah3gpas_pct	263	non-null	int32
44	ah3blue_pct	263	non-null	int32
45	ah3no2_pct	263	non-null	int32
46	ah3so2_pct	263	non-null	int32
47	ah3pm10_pct	263	non-null	int32
48	ah3gamb_pct	263	non-null	int32
49	ah3pubs_pct	263	non-null	int32
50	ah3tob_pct	263	non-null	int32
51	ah3fffood_pct	263	non-null	int32
52	ah3h	263	non-null	float64

```
# count missing values in the AHAH dataset
ahah.isnull().sum()
```

	0
FID	0
LSOA11CD	0
LSOA11NM	0
LSOA11NMW	0
BNG_E	0
...	...
ah3r_pct	0
ah3ahah	0
ah3ahah_rn	0
ah3ahah_pc	0
geometry	0

68 rows × 1 columns

dtype: int64

```
# count Airbnb listings in each LSOA
airbnb.groupby('LSOA11CD').size()
```

	0
LSOA11CD	
E01014485	23
E01014486	45
E01014487	30
E01014488	17
E01014489	51
...	...
E01033364	5
E01033366	10
E01033367	25
E01033369	3
E01033370	4

245 rows × 1 columns

dtype: int64

```
# Save the number of Airbnb listings in each LSOA
airbnb_count = airbnb.groupby('LSOA11CD').size()
```

```
# view Airbnb listing counts
airbnb_count
```

	0
LSOA11CD	
E01014485	23
E01014486	45
E01014487	30
E01014488	17
E01014489	51
...	...
E01033364	5
E01033366	10
E01033367	25
E01033369	3
E01033370	4

245 rows × 1 columns

dtype: int64

```
# convert Airbnb counts to a dataframe
airbnb_count = airbnb_count.reset_index(name='airbnb_count')
```

```
# view Airbnb counts
airbnb_count.head()
```

	LSOA11CD	airbnb_count
0	E01014485	23
1	E01014486	45
2	E01014487	30
3	E01014488	17
4	E01014489	51

```
# merge Airbnb counts with the AHAH dataset
ahah_airbnb = ahah.merge(airbnb_count, how='left', on='LSOA11CD')
```

```
# view the merged dataset
ahah_airbnb.head()
```

	FID	LSOA11CD	LSOA11NM	LSOA11NMW	BNG_E	BNG_N	LONG	LAT	GlobalID	lsoa11	...	ah3r_1
0	14053	E01014485	Bristol 023A	Bristol 023A	360020	174735	-2.57697	51.47015	3edee4ae-aa59-4e9e-8066-fd81222d4d4d	E01014485	...	314
1	14054	E01014486	Bristol 023B	Bristol 023B	359924	174186	-2.57829	51.46521	756ec26e-f1c8-4911-ad30-ff9aab76e295	E01014486	...	358
2	14055	E01014487	Bristol 020A	Bristol 020A	359244	174738	-2.58814	51.47013	3b905c75-f6aa-42c6-8347-f96f779a7b75	E01014487	...	358
3	14056	E01014488	Bristol 023C	Bristol 023C	359701	174621	-2.58155	51.46911	266e675c-152b-4827-abb0-c9424d7b485f	E01014488	...	324
4	14057	E01014489	Bristol 023D	Bristol 023D	359346	174262	-2.58662	51.46585	b608a9d9-8976-4047-b4c5-a5dbf541d1ac	E01014489	...	368

5 rows × 69 columns

```
# check the shape of the merged dataset
ahah_airbnb.shape
```

(263, 69)

```
# replace missing Airbnb counts with 0
ahah_airbnb['airbnb_count'] = ahah_airbnb['airbnb_count'].fillna(0)
```

```
# convert Airbnb counts to integers
ahah_airbnb['airbnb_count'] = ahah_airbnb['airbnb_count'].astype(int)
```

```
ahah_airbnb['airbnb_count'].head()
```

airbnb_count	
0	23
1	45
2	30
3	17
4	51
dtype: int64	

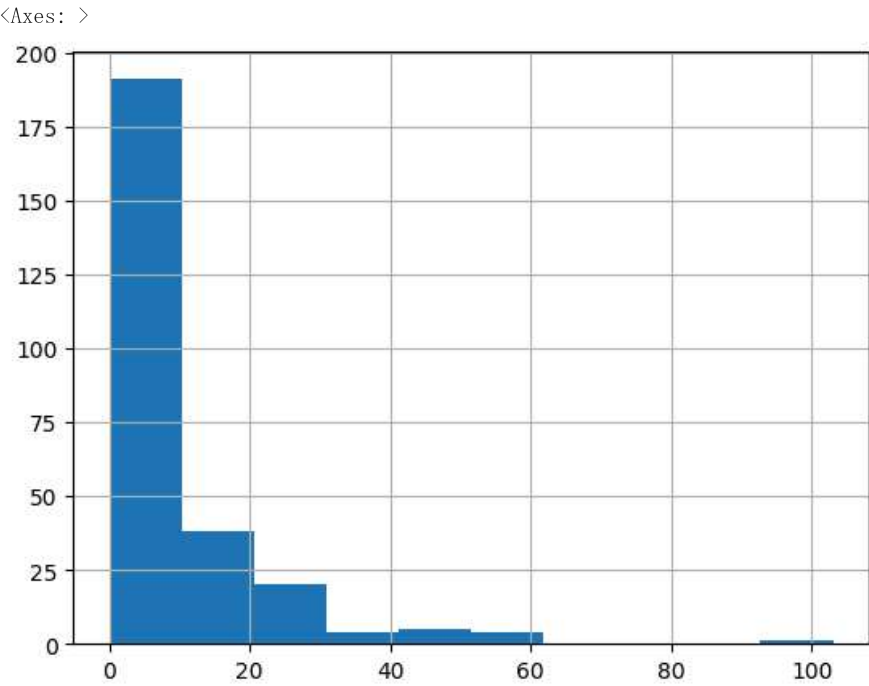
# view the overall AHAH percentile ahah_airbnb['ah3ahah_pc'].head()	
ah3ahah_pc	
0	81
1	77
2	81
3	65
4	86
dtype: int32	

# describe the overall AHAH percentile ahah_airbnb['ah3ahah_pc'].describe()	
ah3ahah_pc	
count	263.000000
mean	59.615970
std	23.921326
min	1.000000
25%	42.500000
50%	62.000000
75%	80.000000
max	99.000000
dtype: float64	

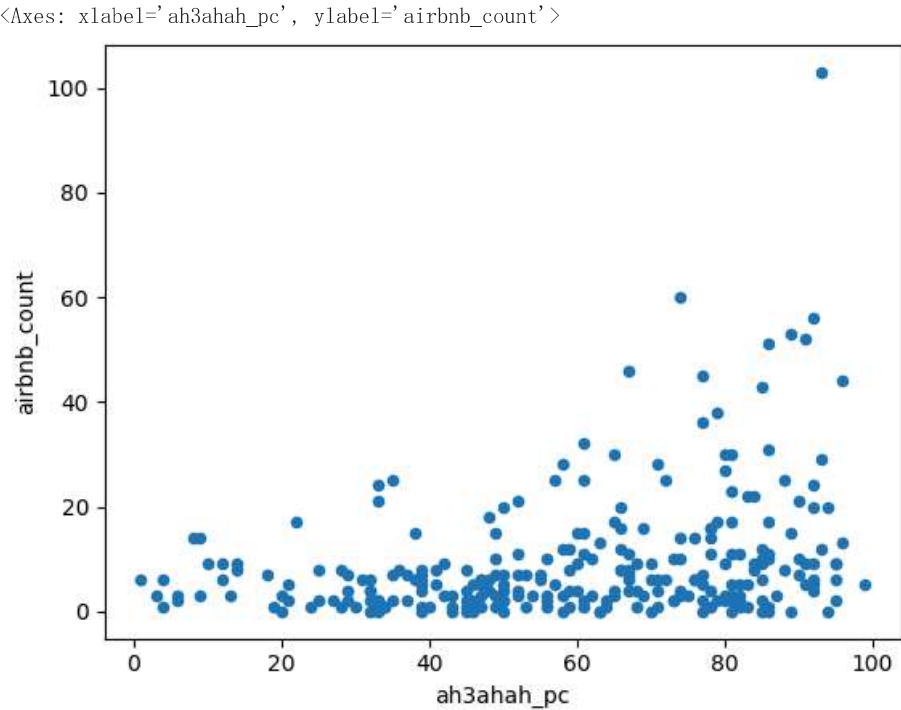
# describe Airbnb listing counts ahah_airbnb['airbnb_count'].describe()	
airbnb_count	
count	263.000000
mean	9.714829
std	12.310799
min	0.000000
25%	2.000000
50%	6.000000
75%	11.000000
max	103.000000
dtype: float64	

--	--

```
# explore the distribution of Airbnb listing counts
ahah_airbnb['airbnb_count'].hist()
```



```
# explore the relationship between Airbnb counts and overall AHAH percentile
ahah_airbnb.plot.scatter(x='ah3ahah_pc', y='airbnb_count')
```



```
# calculate Spearman's rank correlation
ahah_airbnb[['ah3ahah_pc', 'airbnb_count']].corr(method='spearman')
```

	ah3ahah_pc	airbnb_count
ah3ahah_pc	1.00000	0.29762
airbnb_count	0.29762	1.00000

```
# calculate Spearman's rank correlations for the four AHAH domains
ahah_domains_corr = ahah_airbnb[
    ['airbnb_count', 'ah3h_pct', 'ah3g_pct', 'ah3e_pct', 'ah3r_pct']
].corr(method='spearman')

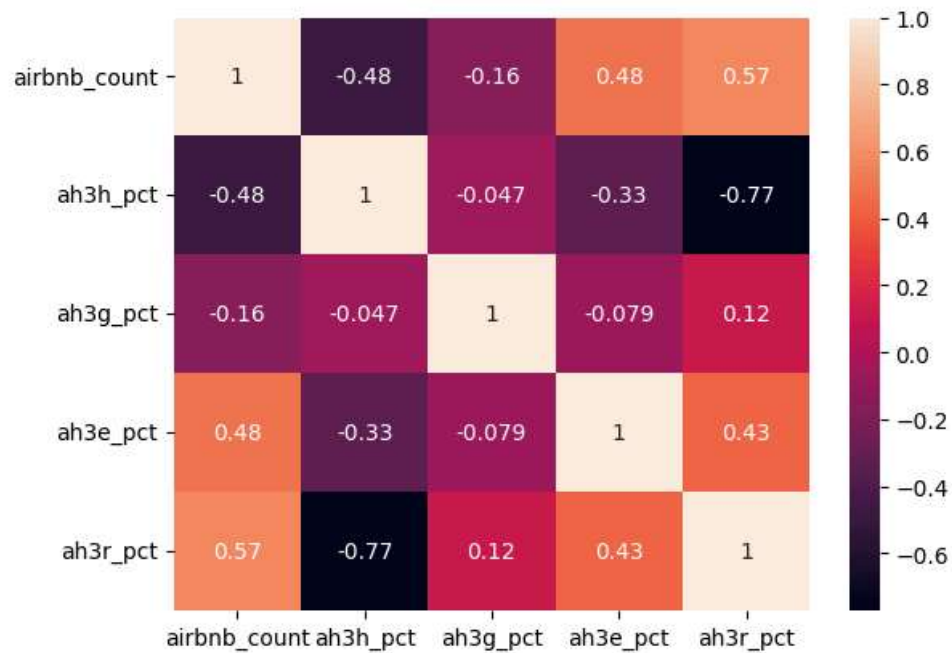
ahah_domains_corr
```

	airbnb_count	ah3h_pct	ah3g_pct	ah3e_pct	ah3r_pct
airbnb_count	1.000000	-0.483697	-0.156235	0.482661	0.565014
ah3h_pct	-0.483697	1.000000	-0.047084	-0.327134	-0.770957
ah3g_pct	-0.156235	-0.047084	1.000000	-0.079417	0.118464
ah3e_pct	0.482661	-0.327134	-0.079417	1.000000	0.427299
ah3r_pct	0.565014	-0.770957	0.118464	0.427299	1.000000

```
# import seaborn for data visualisation
import seaborn as sns
```

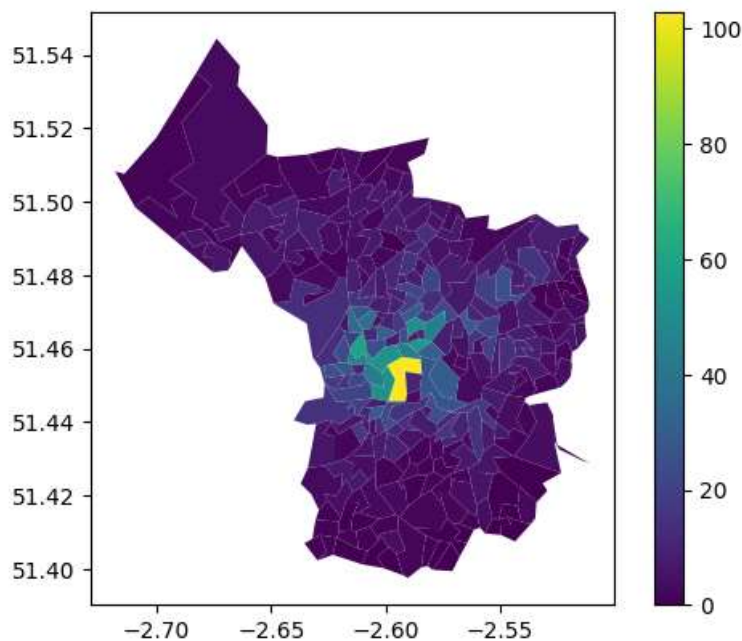
```
# visualise Spearman's rank correlations
sns.heatmap(ahah_domains_corr, annot=True)
```

<Axes: >



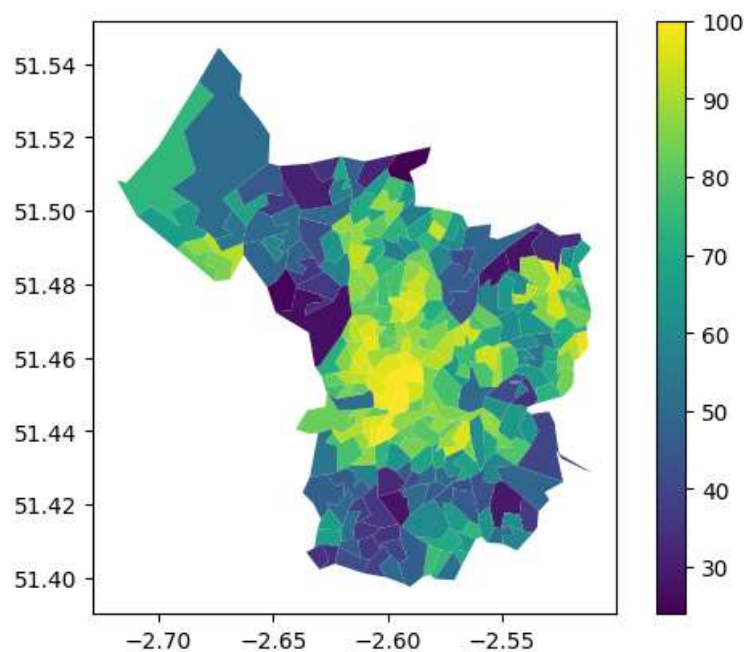
```
# explore the spatial distribution of Airbnb listings
ahah_airbnb.plot(column='airbnb_count', legend=True)
```

<Axes: >



```
# explore the spatial distribution of the Retail Domain percentile
ahah_airbnb.plot(column='ah3r_pct', legend=True)
```

<Axes: >



```
# calculate Spearman's rank correlations for the Retail components
retail_corr = ahah_airbnb[
    ['airbnb_count', 'ah3ffood_pct', 'ah3gamb_pct', 'ah3pubs_pct', 'ah3tob_pct']
].corr(method='spearman')

retail_corr
```

	airbnb_count	ah3ffood_pct	ah3gamb_pct	ah3pubs_pct	ah3tob_pct
airbnb_count	1.000000	0.507771	0.154533	0.719788	0.293404
ah3ffood_pct	0.507771	1.000000	0.568027	0.750033	0.534841
ah3gamb_pct	0.154533	0.568027	1.000000	0.414342	0.298919
ah3pubs_pct	0.719788	0.750033	0.414342	1.000000	0.485478
ah3tob_pct	0.293404	0.534841	0.298919	0.485478	1.000000

```
# select the correlations between Airbnb listings and Retail components
retail_correlations = retail_corr.loc[
    'airbnb_count',
    ['ah3ffood_pct', 'ah3gamb_pct', 'ah3pubs_pct', 'ah3tob_pct']
]

retail_correlations
```

	airbnb_count
ah3ffood_pct	0.507771
ah3gamb_pct	0.154533
ah3pubs_pct	0.719788
ah3tob_pct	0.293404

dtype: float64

```
# rename the Retail components for readability
retail_correlations.index = [
    'Fast food outlets',
    'Gambling outlets',
    'Pubs/Bars/Nightclubs',
    'Vape stores/Tobacconists'
]

retail_correlations
```


	airbnb_count
Fast food outlets	0.507771
Gambling outlets	0.154533
Pubs/Bars/Nightclubs	0.719788
Vape stores/Tobacconists	0.293404
dtype: float64	

2.Methodology

2.1 Data

The study used two sets of data. One was the Airbnb data from Bristol, and the other was the "Access to Healthy Assets and Hazards" (AHAH) data from Bristol.

2.1.1 Airbnb's data

Airbnb is a well-known accommodation platform where individual hosts can list their properties for rent. Tourists can choose from the listings on the platform. Airbnb acts as the intermediary and charges a fee from both parties [2]. The Airbnb data comes from GEOG5990M final project resit project. This dataset contains 28 columns and over 2,500 accommodation information entries. It records the location and basic attributes of the housing units.

2.1.2 The AHAH data

AHAH is the "Index of 'Access to Healthy Assets and Hazards'", which is an index developed by the British Geographical Data Service [6]. This dataset has 68 columns and over 260 pieces of information. The 68 columns of the dataset mainly evaluate the health characteristics of the surrounding environment of each LSOA in four major directions: Retail environment, Health services, Physical environment, and Air quality. The higher the score, the worse the community's health condition.

2.2 Data Preparation and Exploration

By examining the Airbnb data, it was found that there were many missing values. The missing data accounted for 2556 entries, indicating that this content was not included in the overall statistics. It is possible to combine the two datasets with the code 'LSOA11CD', as one of the rows has missing data. Upon checking this row, it was found that it was merely a lack of statistics, not an abnormal data point. This can be disregarded and not included in the calculation among over 2000 data entries.

By examining the AHAH data, it was discovered that this data is intact and has no missing values. It is suitable as an analysis object.

Count the number of Airbnb rentals in each LSOA, obtaining 245 pieces of data. This indicates that there are 18 LSOAs without any rental listings. Combine the two sets of data using LSOA11CD. At this point, the data without missing LSOAs is removed. It cannot be assigned to any LSOA and cannot participate in the regional-level analysis. Therefore, it is excluded during the aggregation process.

LSOAs without matching Airbnb records have missing values. Since these missing values represent that there are no Airbnb listings in those LSOAs, they are not data missing. Replace the missing data in LSOAs without rentals with 0.

2.3 Exploratory and Statistical Analysis

2.3.1 the number of Airbnb listings and the AHAH comprehensive score.

The number of Airbnb listings shows a clearly right-skewed distribution, with a median of 6 and a maximum of 103. This indicates that the majority of LSOAs have a relatively small number of Airbnb listings, while a few regions have a high concentration. For continuous data that does not follow a normal distribution, the Spearman method can be used, and the Spearman method is relatively robust to outliers [7]. Spearman's rank correlation was chosen to study these two sets of data.

Spearman's ρ is approximately 0.3, indicating a positive correlation.

2.3.2

the number of Airbnb listings and the scores in the four directions of AHAH. Retail environment was positively correlated, Air quality was positively correlated, Health services was negatively correlated, while Green/Blue space showed little

relationship.

2.3.3

the number of Airbnb accommodations and the scores in the four aspects of the retail environment. Select Spearman's rank correlation to study these five data. Spearman's ρ indicates that the accessibility of Pubs/Bars/Nightclubs is the most relevant to the number of accommodations, with Spearman's ρ approximately equal to 0.7, followed by the accessibility of Fast food, with Spearman's ρ approximately equal to 0.5. The relationship between the number of Airbnb accommodations and the accessibility to gambling facilities and tobacco stores is not significant. Change the name.

```
import matplotlib.pyplot as plt
```

```
# create the two final visualisations
fig, ax = plt.subplots(1, 2, figsize=(15, 6))

# spatial visualisation
ahah_airbnb.plot(
    ax=ax[0],
    column='airbnb_count',
    legend=True,
    cmap='YlGnBu',
    scheme='quantiles',
    k=5,
    edgecolor='white',
    linewidth=0.2
)
# add a title to the map legend
ax[0].get_legend().set_title('Number of Airbnb listings')

# make the legend labels easier to read
legend = ax[0].get_legend()

labels = [
    '0 - 2',
    '3 - 4',
    '5 - 8',
    '9 - 14',
    '15 - 103'
]

for text, label in zip(legend.get_texts(), labels):
    text.set_text(label)

# map title
ax[0].set_title(
    'Number of Airbnb Listings\n by LSOA in Bristol',
    fontsize=14
)

# remove map axes
ax[0].set_axis_off()

# non-spatial visualisation
bars = ax[1].barh(
    retail_correlations.index,
    retail_correlations.values,
    color=['#7B68EE', '#E76F51', '#2A9D8F', '#E9C46A']
)

# bar chart title
ax[1].set_title(
    'Association between Airbnb Listings\nand AHAH Retail Environment Components',
    fontsize=14
)

# axis label
ax[1].set_xlabel(
    "Spearman's Rank Correlation ( $\rho$ )",
    fontsize=11
)
```

```

# add correlation values to the bars
ax[1].bar_label(
    bars,
    fmt='%.2f',
    padding=3
)

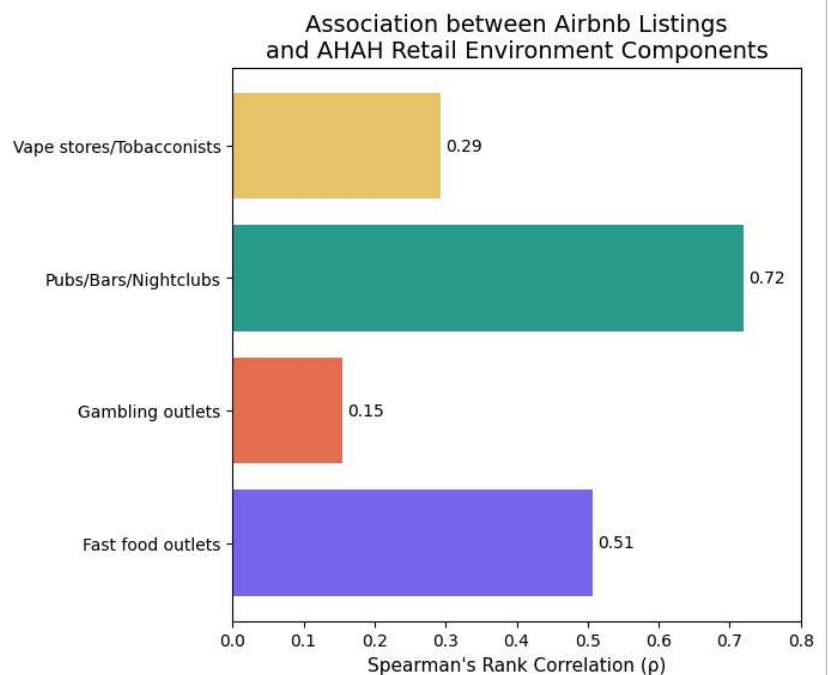
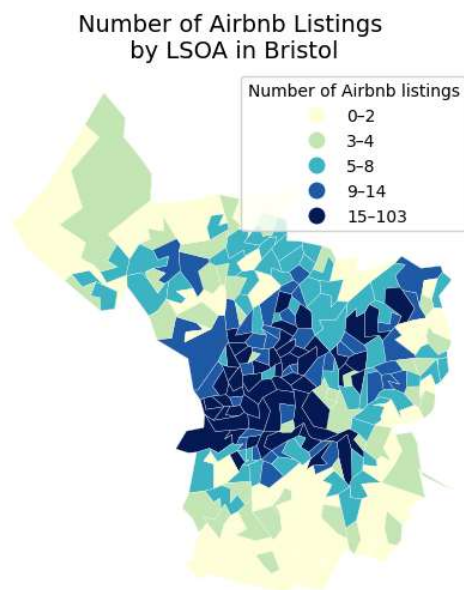
# leave space for the value labels
ax[1].set_xlim(0, 0.8)

# make text easier to read
ax[1].tick_params(axis='y', labels=10)
ax[1].tick_params(axis='x', labels=10)

# add some space between the two visualisations
plt.subplots_adjust(wspace=0.45)

plt.show()

```



#A larger positive Spearman's ρ indicates a stronger positive association between the number of Airbnb list

3.result

3.1 Spatial map

The "YlGnBu" color scheme was selected for the quintile stratified color map. The map shows that the spatial distribution of Airbnb accommodations in Bristol is not uniform. From the map, it can be seen that there are fewer accommodations in the northwest and south areas, and the number of accommodations increases as one gets closer to the city center. Moreover, the accommodations are concentrated in many areas, with a maximum of 103.

3.2Non-spatial chart

From the horizontal bar chart, it can be seen that among the four constituent indicators of the Retail Environment, the number of Pubs/Bars/Nightclubs and Airbnb accommodations shows the strongest positive correlation ($\rho = 0.72$), followed by Fast food outlets ($\rho = 0.51$). According to the definition of AHAH, a higher percentile indicates a relatively less healthy environment in terms of expected health impacts. Therefore, the results suggest that LSOA with a higher number of Airbnb accommodations tend to have higher percentiles in some Retail Environment indicators.

4. Discussion

4.1 Intended Audience

Audience: Airbnb hosts and guests, government departments responsible for tourism, urban planning or short-term rental management

4.2. Visualisation Choices

4.2.1 Spatial map

If the quintile-based layering color map is not set, since the number of accommodations in one of LSOA is much higher than that of others, in the gradient-colored map, only that one LSOA can be seen. The other LSOAs have similar colors because the differences in their values are relatively small. If it is a blue-white gradient, the edge LSOAs will all turn white and have no difference from the background, making it impossible to intuitively see the differences between LSOAs with 0 to 15 accommodations. The quintile classification can make the spatial differences between different LSOAs more easily displayed.

Selecting the "YlGnBu" color scheme, there is a gradual change from light to dark colors, with light colors representing low values and dark colors representing high values. And there are different color changes, the lightest color is not white, and it is different from the background color, making it easier to see clearly.

4.2.2 Non-spatial chart

Because the names of the four categories are too long, if it is a vertical bar chart, the category names should be written slantwise. Choose a horizontal bar chart for display. The colors of the horizontal bar chart are different, and the values can directly indicate. The number of accommodations and the accessibility to bars are positively correlated in the LSOA.

4.3 Limit

The authors who created the AHAH index found that areas with good health environment performance are concentrated in the peripheral areas of the urban core, while areas with poor health environment performance are located in the city center and rural areas [8]. The tourist attractions in Bristol are concentrated in the Old City, Harbourside, and the central area of Clifton [9]. The correlation between Airbnb listings and some AHAH indicators may be influenced by common spatial location factors. These two data are related, but it does not mean a causal relationship.

This study is conducted at the LSOA level. There is still space within an LSOA. Higher Airbnb numbers and higher Pubs/Bars/Nightclubs percentiles occur simultaneously in the same LSOA. This can only indicate an association at the regional level, not that Airbnb accommodations are adjacent to bars.

5. Conclusion

This study investigated the relationship between the number of Airbnb listings in each LSOA of Bristol and the environmental characteristics of AHAH. Airbnb showed a significant spatial concentration. Among them, the Retail Environment had the strongest positive correlation with the number of Airbnb listings, and among the constituent indicators, the correlation of Pubs/Bars/Nightclubs was the highest ($\rho \approx 0.7$). The findings may inform local short-term rental and tourism management. However, the correlation does not imply causation. In the future, more specific analyses can be conducted by including factors such as the distribution of tourist attractions and the distance to the city center that attract tourists.

Do not forget to attach the link to your GitHub repo within your submitted zip file.

References

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